Huawei Access Node BAS-ACC-37 Basavanagudi POP QoS Congestion Event Resolution SOP

Node: BAS-ACC-37 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_2704

Description

The failure domain of QoS congestion events on Huawei Access Node BAS-ACC-37 at Basavanagudi POP can be attributed to the occurrence of microbursts, interface tail drops, and QoS queue starvation. This phenomenon is primarily a result of the interaction between the Access Control Plane and the Data Plane on the Huawei NE40E device. The Access Control Plane, responsible for forwarding decision and policing, may become overwhelmed due to excessive traffic volume, leading to the congestion of QoS queues. The QoS queues, in turn, may experience tail drops, causing packet loss and further exacerbating the congestion. Furthermore, the Huawei NE40E's hardware-based policing mechanism may also contribute to the congestion by dropping packets that exceed the configured policy thresholds. In order to resolve this issue, it is essential to identify and address the root cause of the congestion, which may involve adjusting the policing thresholds, reconfiguring QoS policies, or optimizing the network topology to reduce traffic volume.

Pre-requisite Checks

- Remote hands must be dispatched to the Basavanagudi POP to enable access to the Huawei NE40E device.

- The redundant link status must be verified to ensure that it is functioning properly and can take over in case of a failure.

- The network topology must be reviewed to identify potential bottlenecks or areas where traffic congestion may be occurring.

Potential Root Causes

Cause	Probability
Excessive traffic volume	High
Inadequate policing thresholds	Medium
Congested QoS queues	Medium

Diagnostic Commands & Expected Outputs

```
Command: display interface GigabitEthernet 0/0/0 | include in-use, Expected Output: in-use: yes, Description: This command checks the usage status of the GigabitEthernet 0/0/0 interface and verifies whether it is in use or not.

Command: display qos queue | include queue, Expected Output: queue 1/1 1/1 1/1, Description: This command displays the QoS queue information
```

```
and checks whether the queue is congested or not.

Command: display policing | include policer, Expected Output: policer 1/1 1000000000, Description: This command displays the policing information and checks whether the policer is configured correctly or not.
```

Resolution Steps

Step 1: Step: Adjust policing thresholds, Command: policer 1/1 500000000, Description: This step adjusts the policing threshold to reduce the rate of packet drops and alleviate congestion.

Step 2: Step: Reconfigure QoS policies, Command: qos policy 1/1 priority 1, Description: This step reconfigures the QoS policy to prioritize traffic and reduce congestion.

Step 3: Step: Optimize network topology, Command: display network topology, Description: This step optimizes the network topology to reduce traffic volume and alleviate congestion.

Post-Check Verification

```
Command: display interface GigabitEthernet 0/0/0 | include in-use, Expected Output: in-use: no, Description: This command verifies that the GigabitEthernet 0/0/0 interface is not in use after the resolution steps.

Command: display qos queue | include queue, Expected Output: queue 1/1 0/0 0/0, Description: This command verifies that the QoS queue is no longer congested after the resolution steps.

Command: display policing | include policer, Expected Output: policer 1/1 1000000000, Description: This command verifies that the policer is configured correctly after the resolution steps.
```

Roll-back Procedures

- Step: Restore original policing thresholds, Command: policer 1/1 1000000000, Description: This step restores the original policing threshold to ensure that the network returns to its previous state.

- Step: Undo QoS policy reconfiguration, Command: undo qos policy 1/1 priority 1, Description: This step undoes the QoS policy reconfiguration to ensure that the network returns to its previous state.

- Step: Undo network topology optimization, Command: undo display network topology, Description: This step undoes the network topology optimization to ensure that the network returns to its previous state.